

How to build a statistics bot in almost no time?

A Practical Workflow Using Google Custom Search, OpenWebUI API and Shiny

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chatBotR - Motivation

Why?

- Statistics Austria: **FutureLab** → Proposal was accepted
- Chatbots are powerful but often **hallucinate** or use **non-authoritative** sources
- Many real applications require responses **only from a specific website** or controlled data domain
- Goal: **Prototype** a fast, reproducible workflow to:
 - Restrict responses based to a single web source (e.g, *www.statistik.at*)
 - Keep the architecture flexible (model choice, parameters)
 - Provide a usable demo interface (Shiny)
 - Focus on **rapid experimentation** within the Future Lab project rather than building a fully hardened production system

High-Level Architecture

Pipeline Overview (“*Search + RAG*”)

1. Capture user question
2. **Refine query** using LLM-based optimization
3. **Search** optimized query via Google Custom Search (GCS) restricted to {site}
4. Retrieve *k* top results
5. **Build context** from retrieved pages/documents
6. **Generate answer** with LLM using a dynamic prompt:
 - References **only provided sources**
 - Includes citation instructions and how to format the answer
 - Contains hard-coded fallback answers if no relevant results are found or if the query is non-objective (e.g., “*tell me a joke**”)

Endpoints

Api-Endpoints that were used:

- Google Custom Search:
 - GET customsearch/v1
- OpenWebUI:
 - GET /api/models
 - POST /api/chat/completions
 - POST /api/files/
- Gemini:
 - `ellmer::chat_google_gemini()`

The Shiny-App (UI)

- **Language Selection:** currently *de* | *en* available, easily extendable
- **Parameter Selection**
 - **LLM Host & Model:** Choose backend-server and available models
 - **Google Custom Search (GCS) settings**
 - *max_k*: Maximum number of results passed to LLM
 - **LLM Parameters**
 - *temperature*: Controls randomness/creativity
 - *max_tokens*: Maximum length of generated output
- **Results Display**
 - Shows formatted answer with citations
- **Download Options**

Export content and debug data

The Shiny-App (Server)

- **Configuration:**
 - Based on Env-Vars (e.g. Language, Search Site, API-Keys)
- **Pipeline Orchestration**
 - Executes steps sequentially
 - File-Upload Caching
- **Progress Feedback**
 - Visual progress bar for pipeline stages
- **Error Handling**
 - Pop-up alerts for failed steps

Limitations & Areas for Improvement

- **Latency**

- Multiple LLM calls + web requests take time
- Possible Mitigation steps:
 - **Batching** of requests
 - **Pre-fetching** of data
 - **Benchmark** asynchronous vs. synchronous multi-request patterns for N requests to an LLM API.
 - **Caching** of results

More Areas for Improvement

- **Query Optimization** is model-dependent
 - **Validate or reject queries missing key dimensions** (e.g., region, time)
- **Improving provided context**
 - Current: All top k documents from GCS are passed to LLM
 - Future: **Rank by relevance & recency** (e.g., prioritize latest results)

Try it out

- We have put the code on github.com/statistik.at/chatBotR
- Feedback / Ideas, Suggestions are very welcome!
- We would welcome other contributions (e.g. via PRs) even more :)

Conclusion

Key Takeaways

- Combining GCS + LLM + Shiny enables a powerful, reproducible framework for a domain-restricted chatbot
- The pipeline tries to enforce valid answers with transparent citations
- Modular design allows for quick changes, experiments and easy updates (e.g. adding new models, ..)
- Great platform for further R-based research in retrieval, prompting, and evaluation

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Independent statistics for evidence-based decision making

